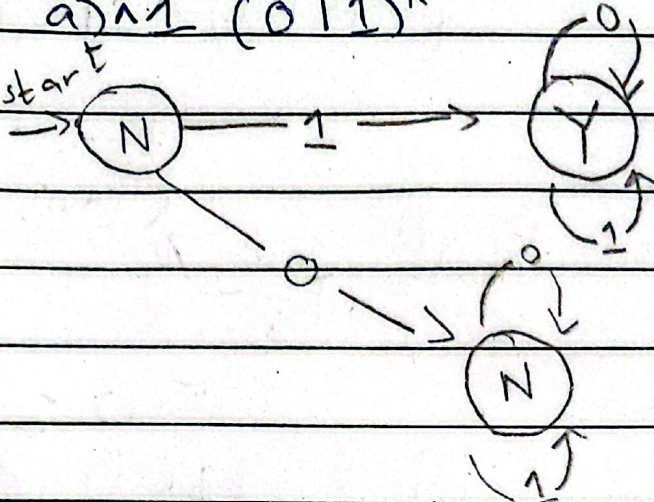


Assignment 4

Date: /

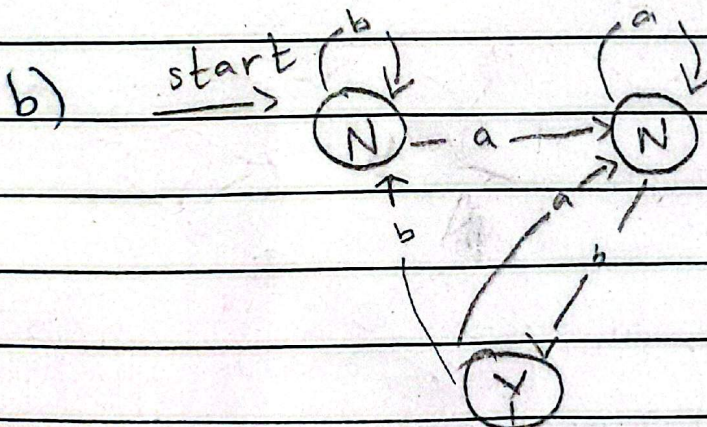
Q1) a) $11(011)^*$

b) start



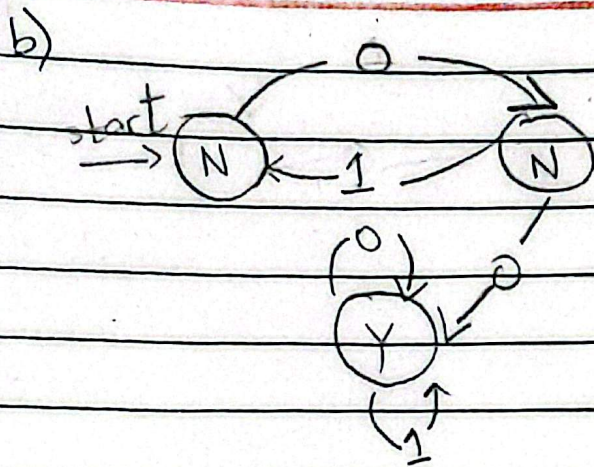
c) Accepted : 1, 100
Rejected : 0, 011

Q2) a) $(a|b)^*ab\$$



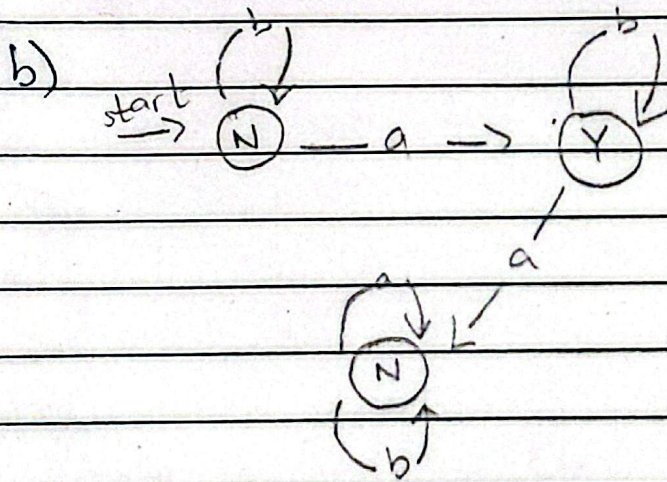
c) Accepted : ab, bbab
Rejected : aba, abb

Q3) a) $(011)^*00(011)^*$



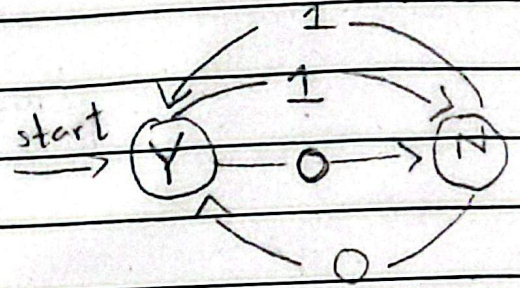
c) Accepted : 00, 1001
Rejected : 010, 111

Q4) $a^*b^*ab^* \$$



c) Accepted : a, bbabb
Rejected : bb, aab

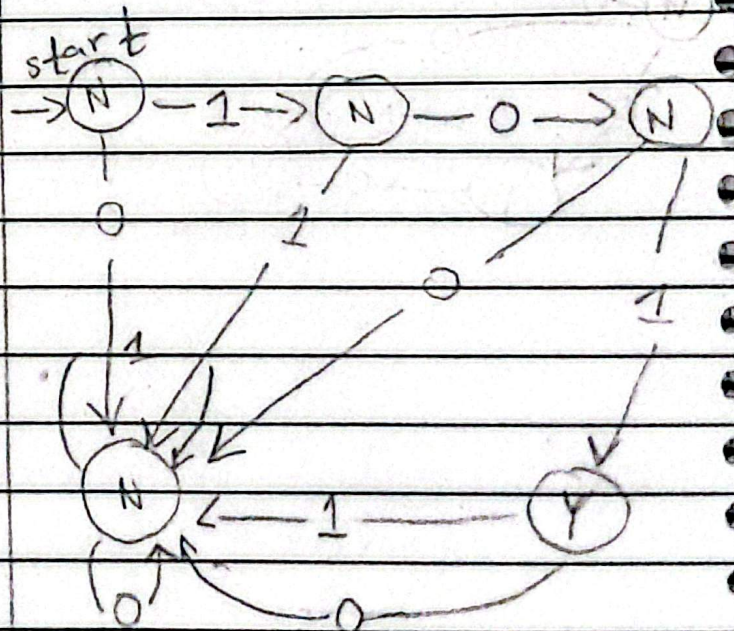
Q5) $a^*(011)(011)^* \$$



Accepted : 1100, 01
Rejected : 0, 101

Q6) $a^*101 \$$

b)



c) Accepted : 101

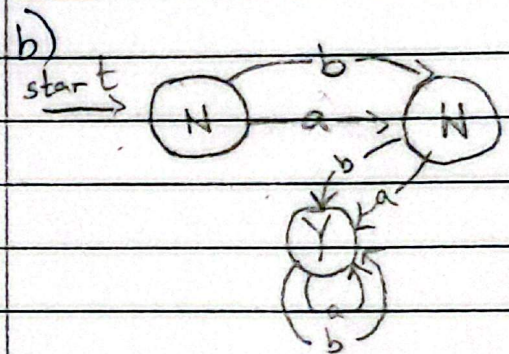
Note: 101 is the only string accepted by this

DFA, since the language consists of exactly one string

Rejected: 10, 1011
~~00000~~

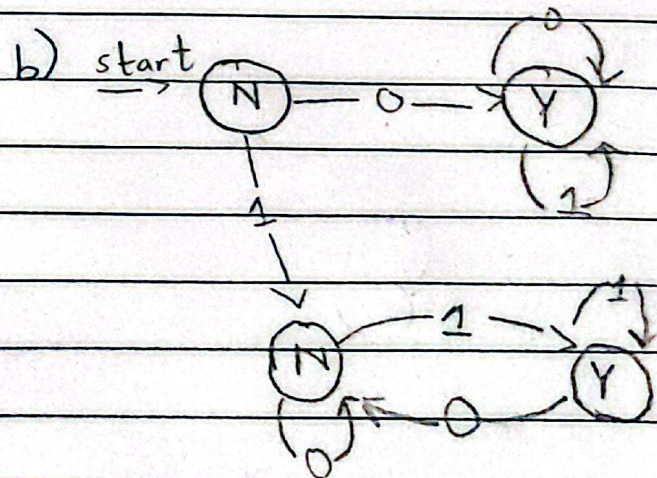
c) Accepted: 1, 1111
Rejected: 10, 01

Q9) a) $(a|b)(a|b)(a|b)^*$

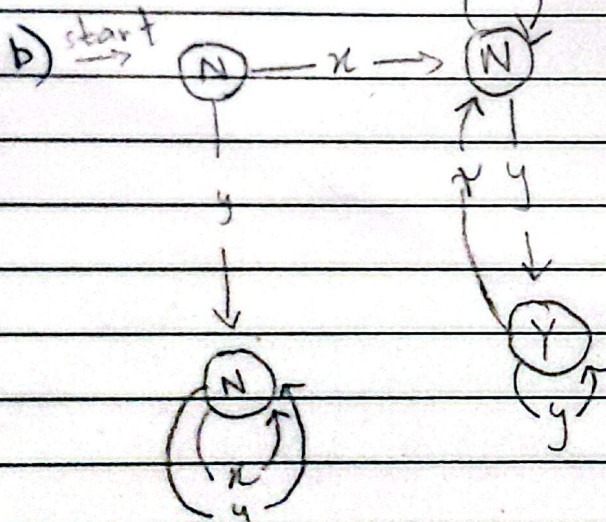


c) Accepted: ab, aab
Rejected: a, " " (empty string)

Q10) a) $^a(0(0|1)^*1(0|1)^*1)^*$
 ~~$(0(0|1)^*1)^*$~~

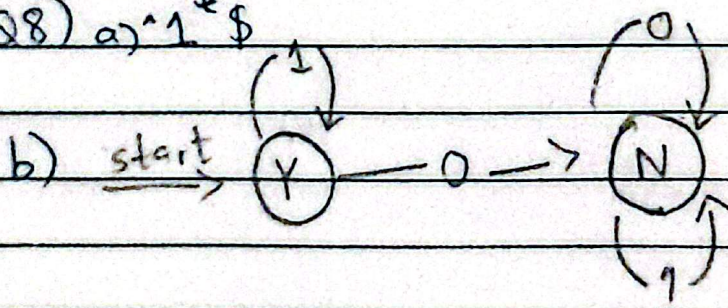


Q7) a) $x^*(x|y)^*y$



Accepted: xy, xxxy
Rejected: yx, xyx

Q8) a) $^a1^*$



c) Accepted : 01, 111
Rejected : 10, 100